

Algoritmos y Estructuras de Datos I

Practico 3

wp condicional o alternativa (**if**)

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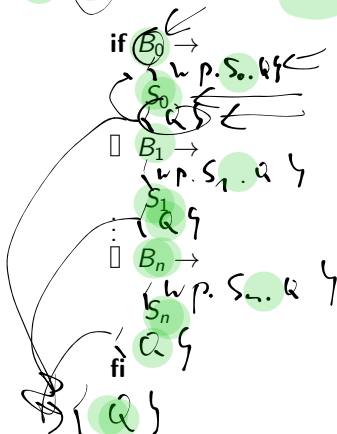
Fa.M.A.F., Universidad Nacional de Córdoba, [Argentina](#)

Precondición más débil:

$$\begin{aligned} wp.(\text{if } \dots \text{fi}).Q &\equiv (B_0 \vee B_1 \vee \dots \vee B_n) \\ &\quad \wedge (B_0 \Rightarrow wp.S_0.Q) \\ &\quad \wedge (B_1 \Rightarrow wp.S_1.Q) \\ &\quad \vdots \\ &\quad \wedge (B_n \Rightarrow wp.S_n.Q) \end{aligned}$$

Precondición más débil:

$$wp.(if \dots fi).Q \equiv (B_0 \vee B_1 \vee \dots \vee B_n) \wedge (B_0 \Rightarrow wp.S_0.Q) \wedge (B_1 \Rightarrow wp.S_1.Q) \dots \wedge (B_n \Rightarrow wp.S_n.Q)$$



Verificación con Precondición más débil:

$\{P\}$ **if** $B_0 \rightarrow S_0$ $\{Q\}$
 $\square$ $B_1 \rightarrow S_1$
 :
 $\square$ $B_n \rightarrow S_n$
fi

$$\equiv P \Rightarrow wp.(\text{if} \dots \text{fi}).Q$$

Verificación con Terna de Hoare:

$$\begin{array}{l} \{P\} \text{ if } B_0 \rightarrow S_0 \\ \quad \square B_1 \rightarrow S_1 \\ \quad \vdots \\ \quad \square B_n \rightarrow S_n \\ \text{fi} \end{array} \{Q\} \equiv P \Rightarrow (B_0 \vee B_1 \vee \dots \vee B_n) \wedge \{B_0 \wedge P\} S_0 \{Q\} \wedge \{B_1 \wedge P\} S_1 \{Q\} \wedge \dots \wedge \{B_n \wedge P\} S_n \{Q\}$$

Programa anotado (con wp):

$\{(B_0 \vee B_1 \vee \dots \vee B_n) \wedge (B_0 \Rightarrow wp.S_0.Q) \wedge \dots \wedge (B_n \Rightarrow wp.S_n.Q)\}$

if $B_0 \rightarrow$

$\{B_0 \wedge wp.S_0.Q\}$

S_0

$\{Q\}$

$\square$ $B_1 \rightarrow$

$\{B_1 \wedge wp.S_1.Q\}$

S_1

$\{Q\}$

$\vdots$

$\square$ $B_n \rightarrow$

$\{B_n \wedge wp.S_n.Q\}$

S_n

$\{Q\}$

fi

$\{Q\}$

Programa anotado (con Hoare):

```
{P}  
if  $B_0 \rightarrow$   
   $\{B_0 \wedge P\}$   
   $S_0$   
   $\{Q\}$   
 $\square$   $B_1 \rightarrow$   
   $\{B_1 \wedge P\}$   
   $S_1$   
   $\{Q\}$   
 $\vdots$   
 $\square$   $B_n \rightarrow$   
   $\{B_n \wedge P\}$   
   $S_n$   
   $\{Q\}$   
fi  
 $\{Q\}$ 
```

Ejemplo

```

Var x: Int;
{ x ≥ -1 }
if x ≥ 0 →
  { x ≥ 0 }
  skip
  { a }
□ x ≤ 0 →
  { x ≤ 0 ∧ x ≥ -1 }
  x := x + 1
  { a }
fi
Q: { x ≥ 0 }
  
```

$$\begin{aligned}
 wp.\text{if} \dots fi. (x \geq 0) &= \\
 & (x \geq 0 \vee x \leq 0) \\
 & \wedge x \geq 0 \Rightarrow wp.\text{skip}.(x \geq 0) \\
 & \wedge x \leq 0 \Rightarrow wp.(x := x + 1).(x \geq 0) \\
 & \equiv \text{True} \wedge x \geq 0 \Rightarrow x \geq 0 \\
 & \wedge x \leq 0 \Rightarrow x + 1 \geq 0 \\
 & \equiv x \geq 0 \vee x \geq -1 \\
 & \equiv x \geq -1
 \end{aligned}$$

Q